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Heterogeneous Multi-Robot Collaboration Using Large Language Models

Exploring different sizes and architectures

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Master's Programme in Embedded Systems

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Abstract

Heterogeneous warehouse fleets typically include robots with different roles (e.g., carrier vehicles and loading agents) that must coordinate their movement, timing, and energy usage to sustain high throughput. Designing controllers for such multi-robot systems is challenging due to large discrete action spaces, partial observability, sparse rewards, and safety constraints such as battery depletion.

This thesis investigates whether off-the-shelf large language models (LLMs) can be used as high-level decision-makers for heterogeneous robot collaboration in a task-assignment warehouse environment with battery dynamics. The approach converts the current multi-agent state into a compact natural-language description, queries an LLM to choose an action for each agent, and enforces feasibility through action-mask validation and rule-based fallbacks. Different model sizes and prompt designs are compared against a heuristic baseline in the Battery-TA-RWARE environment.

The current work contributes (i) an end-to-end evaluation pipeline for LLM-driven multi-agent control, (ii) prompt and validation patterns tailored to discrete warehouse actions, and (iii) preliminary experimental results showing a significant performance gap between a plain-language LLM policy and a domain heuristic, highlighting the need for stronger structure, grounding, and coordination mechanisms.

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Sammanfattning

I heterogena robotflottor i lager används ofta olika robottyper (t.ex. bärande automated guided vehicles (AGVs) och lastningsagenter) som måste samordna sina handlingar för att uppnå hög genomströmning samtidigt som säkerhetskrav, såsom batterinivåer, uppfylls. Att konstruera styrsystem för sådana multi-robot systems (MRSs) är svårt på grund av stora diskreta handlingsrymder, glesa belöningssignaler och behovet av robust koordination.

Detta examensarbete undersöker om färdigtränade LLMs kan användas som beslutsfattare på hög nivå för samarbete mellan heterogena robotar i en lager-miljö med uppgiftsallokering och batteridynamik (Battery-TA-RWARE). Metoden sammanfattar det multi-agentiska tillståndet i naturligt språk, låter en LLM välja en handling per agent och validerar sedan handlingarna mot miljöns tillåtna handlingsmasker, med regelbaserade fallback-strategier när LLM-utdata är ogiltig eller osäker.

Arbetet levererar en första utvärderingspipeline för LLM-styrning av multi-agentisk lagerlogistik samt preliminära resultat som visar att en enkel språkbaserad LLM-policy presterar avsevärt sämre än en domänheuristik, vilket motiverar fortsatt arbete med bättre struktur, grounding och koordination.

Acknowledgement I would like to thank my supervisor, Xuezhi Niu, for continuous guidance and constructive feedback throughout this thesis work. I also thank my reviewer, Didem Gürdür Broo, and my examiner, Ramanathan Thinniyam Srinivasan, for their input on the direction and quality of the work. Finally, I am grateful to the developers and maintainers of the open-source environments and tools used in this project.

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Acronyms

AGV automated guided vehicle. ii, 1, 5, 7–9, 11

JSON JavaScript Object Notation. 12

LLM large language model. i, ii, iv, 1–4, 6–14

MARL multi-agent reinforcement learning. 1, 3, 13

MAS multi-agent system. 4

MRS multi-robot system. ii, 1, 4, 5

PPO proximal policy optimization. 9, 13

RL reinforcement learning. 3, 9

TA-RWARE task-assignment robotic warehouse environment. 7

1 Introduction

Modern warehouses increasingly rely on fleets of autonomous agents to move inventory from storage to packing stations. Even in simplified settings, achieving high throughput requires more than path planning: agents must continuously decide *which* tasks to execute, *when* to coordinate with other roles, and *when* to recharge to avoid downtime. These decisions become harder in heterogeneous teams where agents have different capabilities and constraints.

Research on MRSs spans architectures, communication, and coordination strategies. Early surveys emphasized fundamental axes such as group architecture, resource conflicts, and learning [3]. Later overviews categorized multi-robot research topics including communication, task allocation/control, and motion coordination [2]. In practical logistics domains, these themes manifest as repeated, interdependent choices: allocate requests to carriers, ensure loaders and carriers meet at shelves, resolve conflicts in shared corridors, and maintain energy safety.

Many approaches address such problems using optimization, hand-engineered heuristics, or multi-agent reinforcement learning (MARL). Optimization and heuristics can be strong and efficient, but they often require careful domain-specific design. MARL can learn sophisticated joint behaviour, but training is expensive, sensitive to reward shaping and partial observability, and may generalize poorly when the environment parameters change.

Recent LLMs offer a complementary direction: instead of learning a policy from scratch for each environment, an LLM can be prompted with a structured description of the current state and asked to output a high-level decision. In robotics, LLMs have been used for instruction following and planning by combining language reasoning with explicit grounding and feasibility checks; for example, by selecting from validated affordances rather than unconstrained free-form actions [1]. However, applying LLMs to multi-agent control introduces additional challenges: actions must remain valid under environment constraints, decisions must be coordinated across agents, and long-horizon objectives must be approximated under limited context and latency budgets.

This thesis studies LLM-driven decision-making in a heterogeneous multi-robot warehouse domain with explicit battery dynamics. The experimental platform is Battery-TARWARE, an adaptation of the Task-Assignment Robotic Warehouse environment [5], featuring two agent types: carrier AGVs and picker agents that must meet at shelf locations for load/unload operations. At each decision point, the current global state is summarized into a compact textual prompt, the LLM proposes an action id per agent, and a validation layer enforces action feasibility via action masks and rule-based fall-

backs.

1.1 Research Questions

The work is organized around the following research questions:

1. **RQ1 (Feasibility and safety):** Can an off-the-shelf LLM reliably produce *valid* actions under dynamic constraints (busy states, collisions, role restrictions) while avoiding battery-depletion failures?
2. **RQ2 (Coordination quality):** How well can LLM-based control synchronize heterogeneous roles (carrier–picker meetings) compared to a domain heuristic?
3. **RQ3 (Model and prompt sensitivity):** How do model size/architecture and prompt design choices (candidate action restriction, per-agent vs. shared context, replan triggers) affect throughput and robustness?

1.2 Contributions

The current thesis work contributes:

- An end-to-end pipeline for running and evaluating LLM-driven multi-agent control in Battery-TA-RWARE, including prompt generation, output parsing, and feasibility validation.
- A set of prompt patterns tailored to discrete warehouse actions (requested shelves, goal locations, charging stations) and heterogeneous coordination constraints.
- Preliminary results and an error analysis comparing a plain-language LLM policy against a heuristic baseline.

1.3 Report Outline

Section 2 states the problem formulation, objectives, and delimitations. Section 3 provides background on multi-robot coordination, warehouse task assignment, and LLM-based planning. Section 4 describes the environment and the proposed LLM control loop. Section 5 presents preliminary experiments and discusses observed failure modes. Finally, Section 6 concludes and outlines future work.

2 Problem Formulation and Objectives

2.1 Problem Formulation

Warehouse logistics in heterogeneous fleets can be framed as a continuous decision-making problem: at each time step, each agent must choose a discrete target (shelf, goal, or charging station), subject to dynamic constraints (valid targets, role restrictions, agent busy states) and safety constraints (battery depletion). The objective is to maximize delivery throughput while maintaining robust coordination between roles (carrier–picker interactions) and avoiding failure modes such as deadlocks, clashes, and out-of-charge immobilization.

This thesis investigates LLM-based control as an alternative to purely hand-coded heuristics or fully learned MARL policies. The central question is whether natural-language reasoning can be turned into *actionable* multi-agent decisions once combined with explicit grounding and validation.

2.2 Purpose

The purpose of this thesis is to evaluate the feasibility and limitations of using off-the-shelf LLMs as high-level controllers for heterogeneous multi-robot collaboration in a warehouse task-assignment setting with battery management.

2.3 Objectives

The concrete objectives are:

1. Implement an end-to-end LLM control loop for Battery-TA-RWARE, including prompt generation, output parsing, and feasibility validation.
2. Define an evaluation protocol and metrics that capture throughput, coordination quality, and safety.
3. Compare different LLM sizes/architectures and prompt designs against a heuristic baseline (and, where available, learned reinforcement learning (RL) baselines).
4. Identify and characterize failure modes (invalid actions, poor coordination, battery mismanagement) and propose mitigation strategies.

2.4 Delimitations

To keep the study focused and measurable, the following delimitations apply in the current stage:

- The work is evaluated in a simulated, discrete-time environment (Battery-TARWARE) rather than on physical robots.
- The LLM is used for high-level target selection; low-level navigation is handled by the environment (A* path planning and collision handling).
- The initial LLM policies are *not* fine-tuned; they are prompted as general-purpose models accessed through a local inference server.
- The present results focus on small-scale environments ("tiny") to enable rapid iteration on prompting and validation before scaling to larger warehouses and more agents.

2.5 Ethical and Sustainability Considerations

Although this work is simulation-based, it touches on two practical considerations:

- **Safety:** Policies that ignore constraints (e.g., battery safety) can lead to immobilized agents or unsafe behaviour in real systems. The proposed validation layer is motivated by this gap between natural-language output and executable actions.
- **Energy and compute:** Larger models increase inference cost and energy consumption. A key motivation for comparing different model sizes is to understand whether smaller models can be made effective through better prompting, structure, and validation.

3 Background and Related Work

3.1 Multi-Robot Systems and Coordination

MRSs study how multiple embodied agents share resources, divide work, communicate, and avoid conflicts while pursuing a common objective. The term multi-agent system

(MAS) is broader and can include purely software agents, whereas MRS typically emphasizes physical robots operating under motion, sensing, and actuation constraints. Classical overviews of cooperative robotics highlight recurring concerns such as group architectures, resource conflicts, and learning [3]. In heterogeneous teams, these concerns intensify because roles and capabilities differ, and success often requires explicit synchronization (e.g., meeting at the same physical location at the same time).

In this thesis, *collaboration* refers to interdependent task execution among heterogeneous roles, where agents must coordinate to complete an operation (e.g., a load/unload interaction). *Coordination* refers to mechanisms that prevent conflicts and deadlocks (e.g., collision avoidance and congestion handling), and *cooperation* refers to pursuing a shared objective even when agents may act more independently.

3.2 Warehouse Task Assignment Environments

Warehouse domains provide a structured but challenging testbed for multi-agent decision-making. Tasks are naturally decomposed into repeated cycles: select a requested shelf, bring it to a delivery station, return it to storage, and repeat. When multiple agents operate in the same grid, the action space and coordination requirements scale quickly with environment size and fleet size.

This thesis uses Battery-TA-RWARE, an adaptation of the Task-Assignment Robotic Warehouse environment introduced in [5]. The environment models two heterogeneous agent types:

- **AGVs:** carrier agents that transport shelves between storage locations and goal stations.
- **Pickers:** support agents that must meet AGVs at shelf locations for loading and unloading.

Battery-TA-RWARE extends the task-assignment setting with battery depletion and charging stations, forcing agents to balance throughput with energy safety. Low-level movement is handled by environment dynamics (including A* path planning), while the controller selects discrete targets representing shelves, goals, or charging stations.

3.3 LLM-Based Planning and Grounding

LLMs can produce coherent natural-language plans, but robot control requires converting language into *grounded*, *feasible* actions. One influential pattern is to constrain the LLM to select among validated, environment-defined options. For example, SayCan combines language-level reasoning with an affordance model to filter actions that are executable in the current state [1]. Other work frames LLMs as planners that can propose action sequences given a symbolic interface to the environment, often coupled with explicit checking and replanning [4]. In language-only settings, prompting patterns such as interleaving reasoning and actions have been studied to improve decision quality under tool or environment feedback [6].

These ideas motivate the design in this thesis: use a prompt that provides a compact but explicit mapping from environment state to action identifiers, require outputs to be parseable into discrete actions, and validate them against the environment's current action constraints.

3.4 Why Study Model Size and Prompt Design?

In applied robotics and embedded contexts, compute and latency budgets may favour smaller models. However, smaller models may require more structured prompting, tighter candidate sets, and stronger validation to reach acceptable performance. By evaluating multiple model sizes and prompt variants under the same environment and metrics, the thesis aims to clarify which components (model capacity, prompt structure, validation) most strongly affect feasibility, coordination quality, and throughput.

4 System and Method

4.1 Overview

The thesis uses an empirical, experimental methodology:

1. Implement an LLM-driven control loop for a simulated heterogeneous warehouse environment.
2. Define a set of baselines and evaluation metrics.

3. Run controlled experiments (fixed environment ids and seeds) across multiple LLM models and prompt variants.
4. Analyze throughput, safety, and coordination outcomes as well as qualitative failure modes.

4.2 Environment: Battery-TA-RWARE

Battery-TA-RWARE is a discrete-time, grid-based warehouse simulation derived from task-assignment robotic warehouse environment (TA-RWARE) [5]. At any time, a set of shelves is requested. Carrier AGVs must transport requested shelves to goal locations and then return them to storage. Picker agents are required to meet AGVs at shelf locations for load and unload interactions, creating explicit heterogeneous coordination constraints. Agents additionally have a battery state that decreases with movement and interactions; charging stations are available at fixed locations.

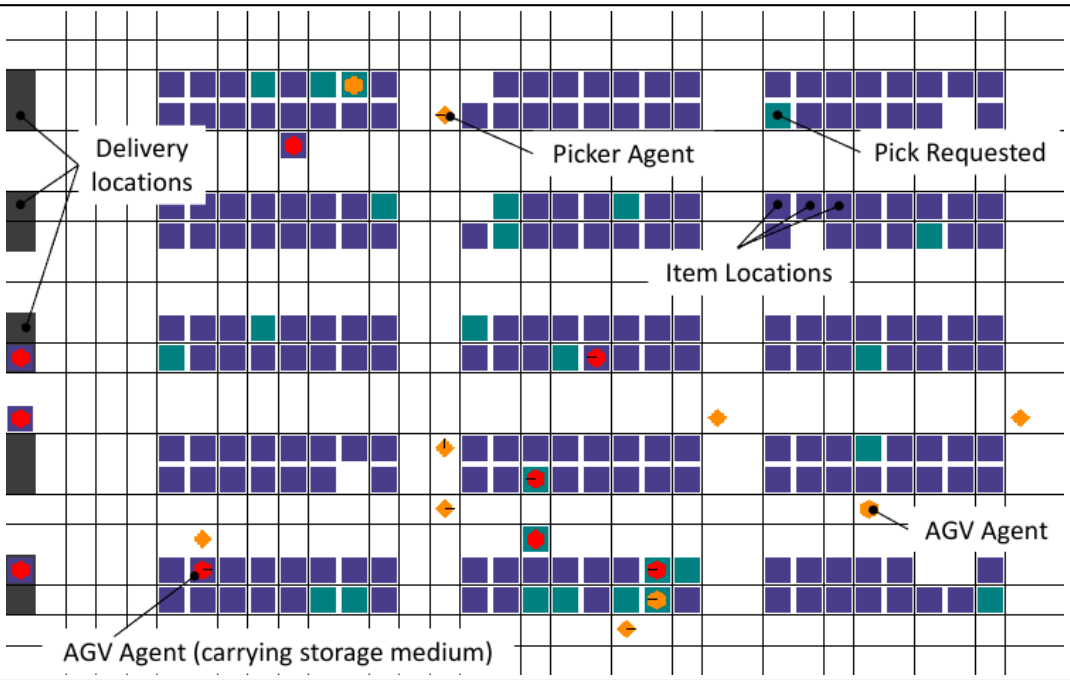

Figure 1 Illustration of the task-assignment warehouse setting used in this thesis. Two heterogeneous roles must coordinate: carrier AGVs transport shelves, while picker agents support loading and unloading at shelf locations. Battery dynamics add a safety constraint that can force agents to recharge.

The environment exposes a discrete action space where each action corresponds to se-

lecting a target location:

- **Shelf actions:** choose a shelf location (e.g., to pick up or return a shelf).
- **Goal actions:** choose a delivery station (primarily relevant when an AGV carries a requested shelf).
- **Charging actions:** choose a charging station to recover battery.

Low-level motion toward the selected target is handled internally by the environment (A* path planning and collision resolution), while the controller is evaluated on the quality of these high-level decisions.

4.3 LLM Control Loop

The LLM policy is implemented as a text-in / discrete-action-out controller:

1. **State summarization:** extract salient features of the global state (requested shelves and their action ids, per-agent position/target/battery/carrying status, charging occupancy).
2. **Candidate action set:** compute a valid-action mask from the environment; optionally restrict the candidate list shown to the LLM to reduce prompt size.
3. **Prompting:** generate a per-agent prompt describing the agent role and the current state, including explicit mappings from requested objects to action ids.
4. **Querying:** send the prompt to a local inference server (Ollama) and obtain a text response.
5. **Parsing:** parse an integer action id (and optionally an action hold duration) from the response.
6. **Validation and fallback:** if the action is invalid under the current mask, replace it with a rule-based fallback action.
7. **Execution:** apply the validated joint action in the environment for one step (or hold the action for multiple steps, depending on configuration).

Two design choices are central for feasibility:

- **Grounding via action ids:** the LLM does not output coordinates or free-form text commands; instead it selects from discrete action ids that are already defined by the environment.
- **Hard validation:** action masks enforce constraints such as invalid goals for picker agents and busy-state restrictions. Invalid LLM outputs are never executed.

4.4 Baselines

The primary baseline is the environment-provided heuristic policy, which assigns agents to requested shelves and deliveries using domain logic and shortest-path distances. Where available, learned RL baselines (e.g., proximal policy optimization (PPO) trained via RLlib) are considered as an additional comparison point, but the current report focuses on heuristic versus LLM as a first-draft milestone.

4.5 Evaluation Metrics

The main metric is delivery throughput, expressed as a pick rate (deliveries per hour) derived from the number of delivered shelves and the simulated episode duration. Additional metrics are collected to capture coordination and safety:

- total shelf deliveries,
- number of clashes and deadlocks/stuck events,
- distance travelled by each agent type,
- episode return (environment reward),
- episode length and inference speed/latency (when measured).

4.6 Experimental Protocol (Current Stage)

To support reproducibility during prompt iteration, experiments are run with fixed environment ids and fixed seeds. The preliminary results in Section 5 are obtained on a small environment configuration ("tiny") with one AGV and one picker agent and a single seed. Subsequent experiments will extend to multiple seeds, additional environment sizes, and larger heterogeneous teams.

5 Preliminary Results and Discussion

5.1 Implementation Status

At the time of writing, the following components are implemented and working end-to-end:

- Battery-TA-RWARE environment integration via Gymnasium.
- Per-agent prompt generation with explicit action-id grounding and role-specific policy hints.
- LLM querying through a local HTTP interface (Ollama).
- Output parsing, action-mask validation, and rule-based fallbacks.
- Episode-level logging and metric aggregation (deliveries, pick rate, clashes, stuck events, return).

5.2 Preliminary Experiment

This section reports a single initial experiment intended to verify the pipeline and establish a first baseline comparison. Configuration:

- Environment: `tarware-tiny-1agvs-1pickers-partialobs-chg-v1`
- Seed: 0
- LLM: `llama3.2:3b` (Ollama)
- Episodes: 1

Metric	LLM policy	Heuristic baseline
Overall pick rate (deliveries/hour)	2.08	10.12
Total shelf deliveries	1	20
Global episode return	-3.44	17.53
Total clashes	30	88
Total stuck events	6	14
Episode length (steps)	346	1423

Table 1 Preliminary comparison between the LLM-driven controller and the built-in heuristic baseline on a tiny environment with two agents. These numbers are from a single seed and are not yet statistically representative.

The LLM policy achieves substantially lower throughput than the heuristic on this initial run. The LLM episode also terminates earlier, which reduces the opportunity to accumulate deliveries. While the LLM policy produces fewer clashes in absolute terms, this is consistent with lower overall activity rather than better coordinated motion.

5.3 Observed Failure Modes

Qualitative inspection of logs suggests the following recurring issues:

- **Weak long-horizon consistency:** the LLM may switch targets frequently or select targets that do not contribute to completing the load–deliver–return cycle.
- **Role/phase confusion:** correct decisions depend on the current phase (e.g., AGV carrying requested shelf should go to a goal; AGV carrying an already delivered shelf must return it to an empty slot). Even with prompt hints, mistakes occur without stronger structure.
- **Prompt overload:** listing large candidate action sets can dilute attention. Restricting candidate actions shown to the model may improve performance but risks hiding necessary options.
- **Battery myopia:** the policy sometimes delays charging too long or charges too early, both of which harm throughput.

5.4 Discussion

The preliminary result supports the hypothesis that plain natural-language prompting is insufficient to match a domain heuristic in a constrained, discrete, multi-agent environment. However, the experiment also validates the feasibility of the architecture in Section 4.3: grounding decisions to action ids, validating them against action masks, and falling back to safe actions prevents catastrophic invalid executions.

The next iterations focus on improving LLM decision quality without sacrificing feasibility. Planned improvements include stricter output formats (e.g., JavaScript Object Notation (JSON)), tighter per-role candidate sets, event-driven replanning triggers, and richer state features that explicitly encode task phase and required coordination steps.

6 Conclusions and Future Work

This thesis explores LLM-driven decision-making for heterogeneous multi-robot collaboration in a task-assignment warehouse setting with battery constraints. The work implemented an end-to-end control loop that (i) grounds decisions to discrete environment action identifiers, (ii) queries an LLM using role-specific prompts, and (iii) enforces feasibility through action-mask validation and safe fallback actions.

The preliminary evaluation shows that a plain-language LLM controller performs substantially worse than a domain heuristic in throughput on a small-scale environment, despite the feasibility layer preventing invalid execution. These results indicate that, in this domain, LLM reasoning alone is not sufficient: performance depends critically on structure (prompt design and output format), grounding (clear state-to-action mappings), and explicit coordination mechanisms.

Nevertheless, the implemented pipeline provides a concrete platform for systematically studying how model size and prompt design affect multi-agent feasibility, safety, and collaboration quality, and it establishes a baseline for future improvements and comparisons.

6.1 Future Work

- **Structured outputs and tighter parsing:** require the model to output a strict schema (e.g., {`action_id, hold_steps, rationale`}) and reject non-conforming responses to reduce ambiguity.

- **Centralized versus decentralized planning:** compare per-agent prompts (current approach) with a centralized planner that outputs a joint action vector, including explicit coordination constraints.
- **Candidate-set design:** learn or compute compact candidate action sets (e.g., nearest requested shelves, relevant support targets for pickers) while guaranteeing that necessary actions remain available.
- **Multi-seed and scaling studies:** extend evaluation to many random seeds and larger layouts (more shelves, more agents) to quantify robustness and scalability.
- **MARL baselines:** train/evaluate PPO or other MARL methods on Battery-TARWARE and compare to LLM in throughput, generalization, and compute cost.
- **Cost-aware deployment:** measure inference latency, token usage, and energy cost across model sizes to assess feasibility for embedded or real-time settings.

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A Appendix

A.1 Reproducibility: Running the Controllers

The experiments in this thesis are executed from the Battery-TA-RWARE repository in this workspace. Example commands:

```
python Battery-TA-RWARE/scripts/run_llm.py \
  --env_id tarware-tiny-1agvs-1pickers-partialobs-chg-v1 \
  --seed 0 \
  --num_episodes 1 \
  --model llama3.2:3b
```

Listing 1: Run the LLM controller for one episode.

```
python Battery-TA-RWARE/scripts/run_heuristic.py \
  --env_id tarware-tiny-1agvs-1pickers-partialobs-chg-v1 \
  --seed 0 \
  --num_episodes 1
```

Listing 2: Run the heuristic baseline for one episode.

A.2 Prompt Excerpt (Illustrative)

The LLM is prompted with an explicit mapping from relevant state elements to discrete action identifiers. The full prompt is generated programmatically; the excerpt below illustrates the format (line breaks and ordering may vary):

```
Current global state:
- Requested shelves:
  - Requested shelf 24 is at position (11,5). To go there, choose
    action id 50.
  - Requested shelf 39 is at position (6,8). To go there, choose
    action id 29.
...
You are agent_0 (type=AGV).
Choose exactly one action_id from the candidate list.
Output format: action_id=<integer>
```

Listing 3: Illustrative prompt excerpt.

A.3 Metric Definitions

- **Total shelf deliveries:** number of requested shelves delivered to goal locations during an episode.
- **Overall pick rate:** deliveries/hour computed as

$$\text{pick_rate} = \frac{\text{deliveries} \cdot 3600}{5 \cdot \text{episode_steps}},$$

  where 5 steps/second is the environment's time scale used by the logging script.
- **Clashes and stuck events:** environment-defined counts related to collision resolution and deadlock-like behaviour.
- **Global episode return:** sum of environment rewards across all agents and time steps.